

MUBASHIR TANWAR

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EXPERIENCE

SDE-1 @ *Prana (PranaIndia)*

July 2025 – Present, *Bangalore*

- Built the customer-facing [pranaindia.com](#) (Next.js/TypeScript) and the **real-time chat and voice-call** consultation service **from scratch**, buffering messages in **Redis** and **batch-flushing** them to the database, sustaining **150+ concurrent chats at sub-200ms latency** with AES-GCM encrypted messaging.
- Built a TypeScript/Express **microservices** backend, deployed on **AWS EC2 with auto-scaling**, ECR, and Amplify using **blue-green deployments**.
- Built an in-house **A/B testing and user-behaviour tracking library**, running **20+ experiments across 150,000+ weekly users** to drive strategic changes that lifted landing-to-consultation conversion **3x (2% → 6%)**.
- Shipped **office.pranaindia.com**, a dedicated portal for high-ticket clients, and streamlined existing consultation workflows.

Software Developer Intern @ *ORIM Advisors*

Jan 2025 – May 2025, *Mumbai*

- Built the front-end console for an internal cloud platform in **React/TypeScript**, using TanStack Query and Zustand for server- and client-state management.
- Led an in-house, **accessible component library** adopted across multiple internal services, cutting **~2 dev-days/week** of duplicated UI work.
- Contributed to the **system design** of core cloud infrastructure modelled on **S3, EC2, and RDS**.
- Built **scraping pipelines**, internal utility APIs, and custom trading-data UIs used across company systems.

Deep Learning Intern @ *National Informatics Centre*

Jan 2024 – Oct 2024, *Mumbai*

- Fine-tuned a **TrOCR Vision Encoder-Decoder** (Google ViT encoder + Marathi-BERT-v2 decoder) in PyTorch/Hugging Face to **83% accuracy**, used to digitize handwritten Marathi government letters received by the office.
- Automated 90%** of dataset labelling to build a self-curated corpus of **135,000+** line- and word-level image-text pairs.
- Built a production **Django API** for PDF OCR with **Elasticsearch**-backed document deduplication, which passed a successful alpha test in the NIC Mumbai office.

Full Stack Engineer @ *Fuzzy Nodes*

Feb 2024 – Sept 2024, *Remote*

- Rebuilt the bestfares365 CRM and airline-booking platform on the **MERN stack**, integrating the **Amadeus API** for flight search, fares, and ticketing.
- Implemented a **fare-search algorithm** layered on Amadeus that widened the search space across route and connection combinations and sorted candidates to **surface cheaper itineraries** than direct queries.
- Built an **agent wallet** with transaction logs and a **dynamic-pricing engine** applying configurable markups per route, airline, and agent.
- Replaced the company's legacy booking system with the new platform, adopted across both customer and travel-agent channels.

PROJECTS

CoWordle — Real-time Multiplayer Wordle — [GitHub](#)

Next.js, Socket.IO, MongoDB

Real-time multiplayer Wordle ([live](#)) where players in a room race on the same word over Socket.IO. The server validates guesses and holds the answer so clients can't cheat, and reconnects restore in-progress games. MongoDB stores rooms, rounds, and scores.

Sudoku — Solver / Generator — [GitHub](#)

C++

Benchmarked **backtracking** against **Knuth's Dancing Links** (Algorithm X over a 324-constraint exact-cover matrix). On a puzzle built to break backtracking, DLX visits **81 nodes vs 69 million (0.03ms vs 1.9s)**. The generator seeds the diagonal boxes and uses DLX to guarantee every puzzle has a **unique solution**.

Timetable Generation System (UCTP) — [GitHub](#)

Python

Built a **Genetic Algorithm** solver for the University Course Timetabling Problem (UCTP), modeling room, instructor, and student-group clashes as hard constraints and preferences and gaps as soft, and tuning a penalty-based fitness function to converge on conflict-free schedules.

EDUCATION

M.H. Saboo Siddik College of Engineering (MHSSCE), University of Mumbai

2021 – 2025

Bachelor of Engineering in Computer Science Engineering (AIML), CGPA: 8.0/10.0

SKILLS

Languages: C, C++, Python, JavaScript, TypeScript, Golang
Frontend: React.js, Next.js, Tailwind, Redux, Tanstack
Backend: Node.js, Express, FastAPI, Prisma
Tools: Git, Postman, Vercel, Notion

Databases: MongoDB, MySQL, Postgres, Redis
Libraries: Numpy, Pandas, OpenCV
DevOps: AWS (EC2, S3, ECR), Docker